WEEK 7 - HEMATOLOGY • LECTURE 8 • LECTURE SLIDE COMPANION

Myeloid Neoplasms

Natasha M. Savage, MD, FCAP • designed to follow alongside the 44-slide lecture deck

Official Week 7 LOs | Slide-order companion | 4 ScholarRx Bricks | Morphology + genetics | Faculty gap-fills labeled | Factoid sheet companion

Overview: Official Week 7 Learning Objectives

LO 1. Distinguish acute from chronic myeloid neoplasms based on cellular differentiation, tempo, and typical clinical consequences.

LO 2. Recognize major AML categories conceptually, including blasts and key distinguishing features, and understand that genetic abnormalities influence prognosis and therapy without requiring detailed classification.

LO 3. Explain acute promyelocytic leukemia (APL): PML-RARA, DIC, and the rationale for differentiation therapy with ATRA.

LO 4. Have a basic understanding of myelodysplastic syndromes, including tempo, characteristic blood-count abnormalities, and typical clinical consequences.

LO 5. Describe CML as a prototype of targeted therapy, including BCR-ABL and constitutive tyrosine kinase signaling.

LO 6. Identify the core molecular drivers and key clinical features of PV, PMF, and ET, emphasizing JAK-STAT dysregulation, blood-count abnormalities, thrombosis, marrow fibrosis, and splenomegaly.

SCHOLARRX BRICKS PAIRED WITH THIS LECTURE

Chronic Myeloproliferative Disorders: Foundations & Frameworks
Acute Myeloid Leukemia
Chronic Myeloid Leukemia
Polycythemia Vera

LO 1

Slides 4-8: myeloid lineage, blasts, acute vs chronic, AML/MDS/MPN/MDS-MPN framework.

LO 2-3

Slides 9-16: AML marrow failure, Auer rods/genetics, APL morphology, PML-RARA, DIC, ATRA, differentiation syndrome.

LO 4

Slides 18-24: ineffective hematopoiesis, cytopenias, dysplasia, pseudo-Pelger-Huet cells, ring sideroblasts, risk scoring.

LO 5

Slides 29-34: CML blood/marrow pattern, BCR-ABL1, Philadelphia chromosome, low LAP, imatinib and molecular monitoring.

LO 6

Slides 35-39: PV, ET, PMF - JAK2/CALR/MPL, dominant cell line, complications, and morphology.

SOURCE HIERARCHY

The **44-slide Savage deck** is the primary slide-order source. The uploaded “Lecture 4 and 5” companion says it was based on another 2026 version and adds treatment, CMML, and board-style distinctions. I use those additions only when they help the official objectives and label them as companion enrichment. The official LO sheet is the authority for what deserves emphasis.

THE ONE FRAMEWORK THAT MAKES THIS LECTURE MANAGEABLE

AML = too many immature blasts.

MDS = cells are made badly and die / fail to exit marrow.

MPN = too many mature myeloid cells are produced.

MDS/MPN = dysplasia + proliferation together.

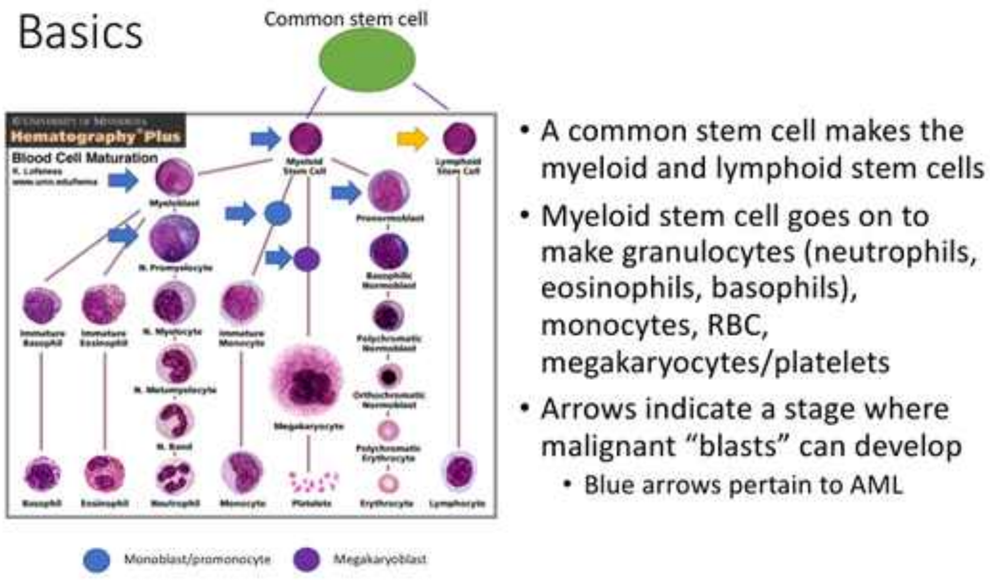
Slides 4-8

Definitions, lineage, and acute vs chronic disease

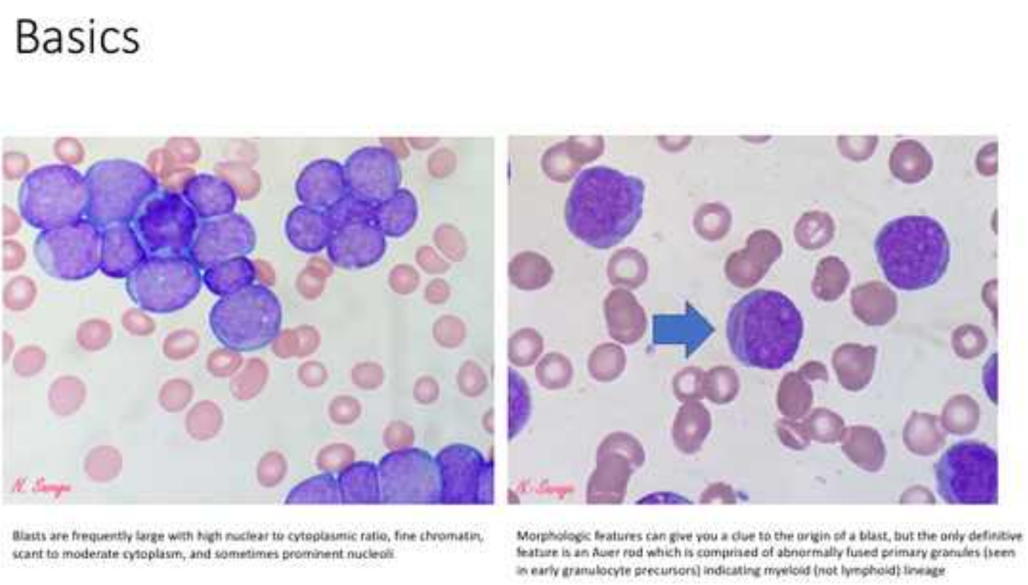
LO 1

Slide 4: define the words before memorizing diseases

- **Myeloid** includes granulocytes, monocytes, erythroid precursors, and megakaryocytes/platelets - everything hematopoietic outside the lymphoid lineage.
- **Acute leukemia** is blast-rich, rapidly progressive, and fatal without prompt treatment.
- **Chronic leukemia** contains predominantly more mature cells and usually has a more indolent course.
- A myeloid blast mass outside marrow/blood is called a **myeloid sarcoma**.



Lecture slide 5 - full slide retained. The blue/purple arrows show stages at which malignant myeloid blasts can arise. Use this to remember that AML is not only a neutrophil-precursor disease; monocytic, megakaryocytic, erythroid, and other myeloid blast phenotypes exist.



Lecture slide 6 - full slide retained. Blasts have high nuclear-to-cytoplasmic ratio, fine chromatin, and sometimes nucleoli. Savage emphasizes that an Auer rod - fused primary granules - is definitive evidence of myeloid rather than lymphoid lineage.

NEED TO MEMORIZE

Auer rod = myeloid.

It is essentially a needle-like bundle of abnormal primary granules and is highly useful when morphology is otherwise ambiguous.

ACUTE VS CHRONIC IS ABOUT DIFFERENTIATION + TEMPO

The key difference is not simply “high WBC vs low WBC.” Acute leukemia is dominated by **immature blasts that crowd out normal marrow**. Chronic disease is dominated by more mature malignant cells that may circulate in enormous numbers but retain more differentiation.

LO 1 checkpoint

Can you define a blast, state what an Auer rod tells you, and explain why a chronic myeloid neoplasm can have a huge WBC yet still be “chronic”?

Slides 9-16

AML and the APL emergency

LO 2 + LO 3

Slide 9: AML in general

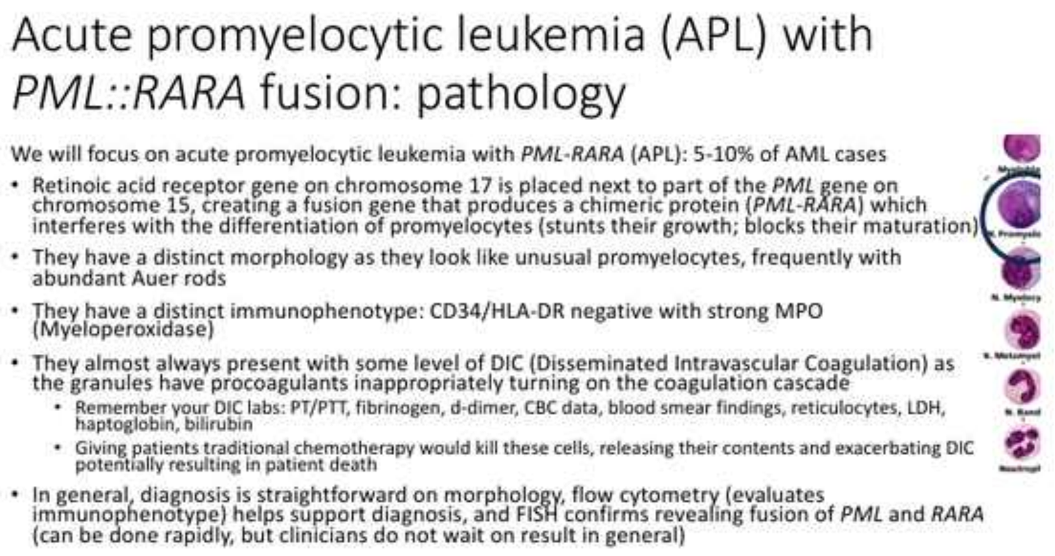
- Classic threshold: **20% or more myeloid blasts/blast equivalents** in blood and/or marrow.
- Savage notes important genetic exceptions in which the defining cytogenetic abnormality can establish AML below 20% blasts.
- Marrow crowding produces the clinical triad: **anemia -> fatigue/pallor; thrombocytopenia -> bleeding/bruising; neutropenia -> infection.**
- Modern WHO/ICC classification uses morphology + immunophenotype + cytogenetic/molecular information.

OFFICIAL LO SAYS CONCEPTUAL - DO NOT DROWN IN CLASSIFICATION TABLES

The official objective explicitly says you should appreciate that genetics changes prognosis and therapy **without requiring detailed classification**. Savage also labels the detailed WHO/ICC MDS tables “No need to memorize these.” Learn the recurring genetic fingerprints, not every line of the classification.

AML genetic clue	Lecture association	What to remember
t(8;21)	Subset of AML with maturation	Generally favorable prognosis in the lecture.
inv(16)	Myelomonocytic AML with abnormal eosinophils	Harlequin eosinophils; favorable; monocytic disease can infiltrate gums/tissues.
KMT2A/MLL 11q23	Often monocytic differentiation	Poor prognosis in the lecture; gingival/organ/tissue involvement can occur.
PML-RARA	APL	**DIC emergency + differentiation therapy.**

Acute promyelocytic leukemia (APL)



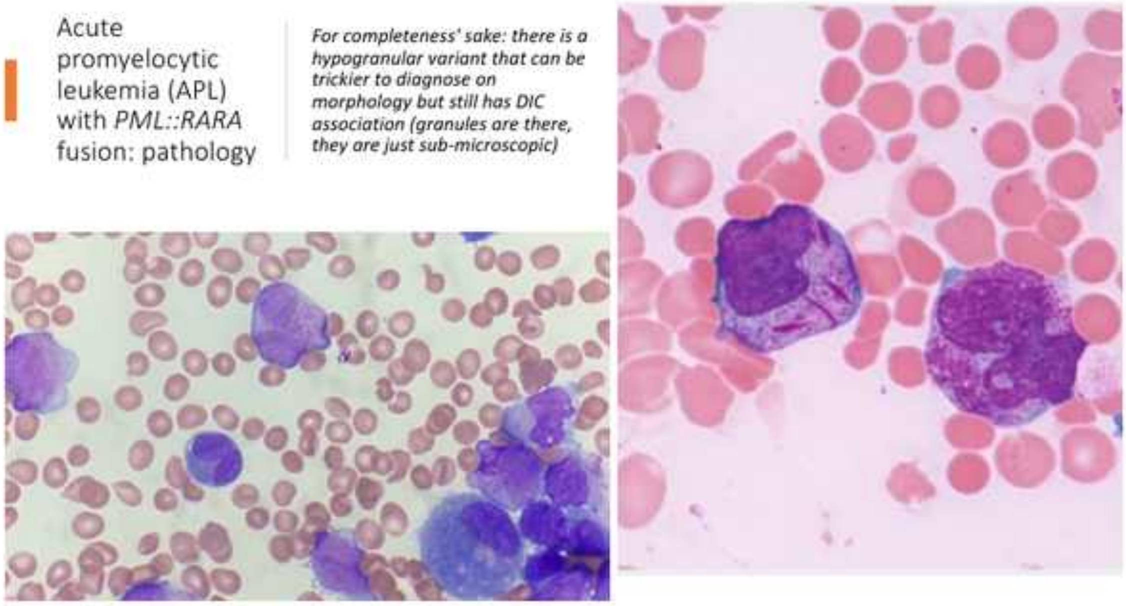
Lecture slide 11 - full slide retained. APL is a maturation block at the promyelocyte stage. The slide emphasizes abnormal promyelocytes with abundant Auer rods, strong MPO, CD34/HLA-DR negativity, and a very strong DIC association.

APL COLD SET

t(15;17) / PML-RARA -> abnormal promyelocytes -> DIC -> start ATRA immediately when suspected.

MECHANISM OF ATRA

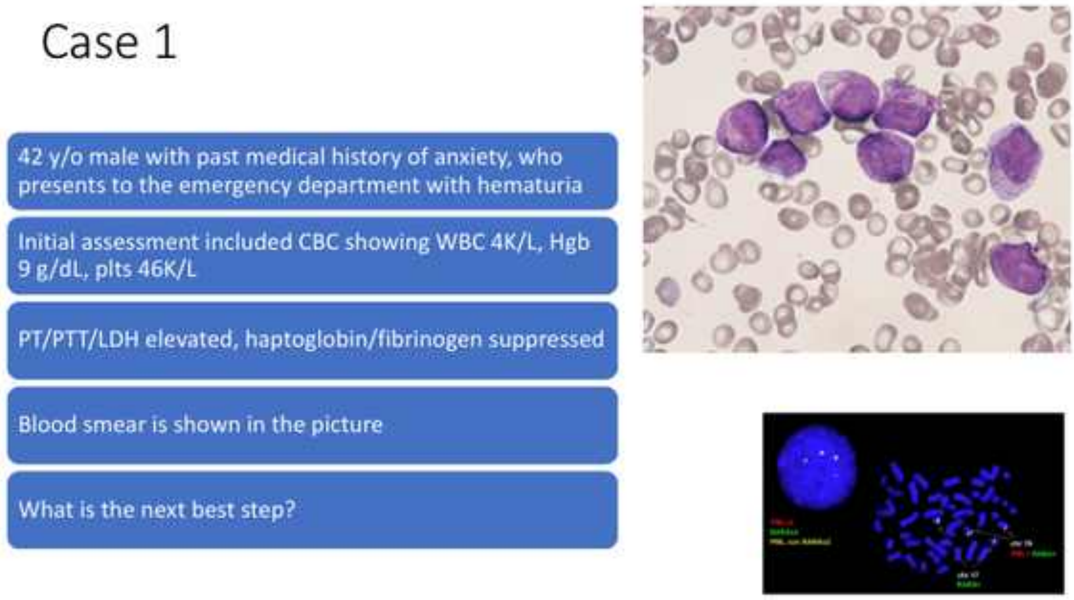
PML-RARA blocks normal retinoic-acid receptor signaling and arrests differentiation. **ATRA binds the RARA portion and relieves/degrades the differentiation block**, allowing malignant promyelocytes to mature instead of simply being lysed en masse.



Lecture slide 12 - full slide retained. The hypogranular APL variant can look less obviously granulated on routine microscopy, but the DIC risk remains because the procoagulant granules are still present at a submicroscopic level.

WHY THE LECTURE SAYS TIME IS OF THE ESSENCE

Savage explicitly says to **start ATRA before confirmatory FISH returns** when APL is strongly suspected. The immediate cause of death is often hemorrhage/DIC, so waiting for perfect certainty is dangerous.



Lecture slide 14 - full slide retained. The case combines hematuria, thrombocytopenia, anemia, prolonged PT/PTT, low fibrinogen, and abnormal promyelocytes. The intended action is immediate ATRA while rapidly confirming PML-RARA.

Slide 15: differentiation syndrome

- Occurs days to weeks after differentiation therapy as maturing myeloid cells release cytokines.
- Think **dyspnea/pulmonary edema, weight gain/edema, fever, hypotension, rising WBC, renal/liver abnormalities, pleural/pericardial effusions.**
- Lecture treatment: **dexamethasone**; severe cases may require holding ATRA/ATO.

LO 2-3 checkpoint

Can you recognize AML from marrow-failure symptoms + blasts, then recognize APL specifically from abnormal promyelocytes/Auer rods + DIC and state why ATRA is started immediately?

Slides 18-24 Myelodysplastic syndromes: ineffective hematopoiesis LO 4

Slide 19: the paradox is the diagnosis

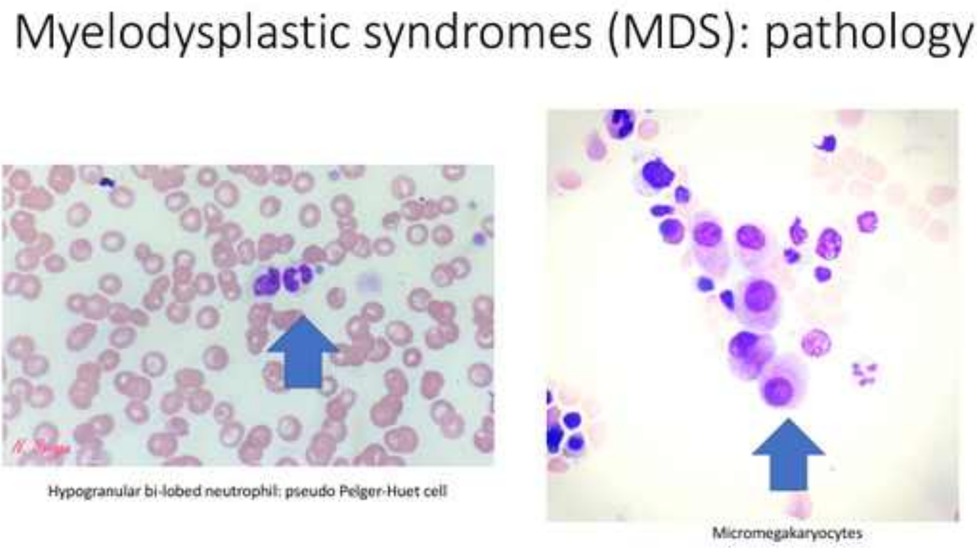
MDS is a clonal stem-cell disorder in which marrow cells are **dysplastic and ineffective**. The marrow can be hypercellular, yet the peripheral blood shows **cytopenias** because abnormal cells fail to mature/exit or die in the marrow.

MDS ONE-LINER

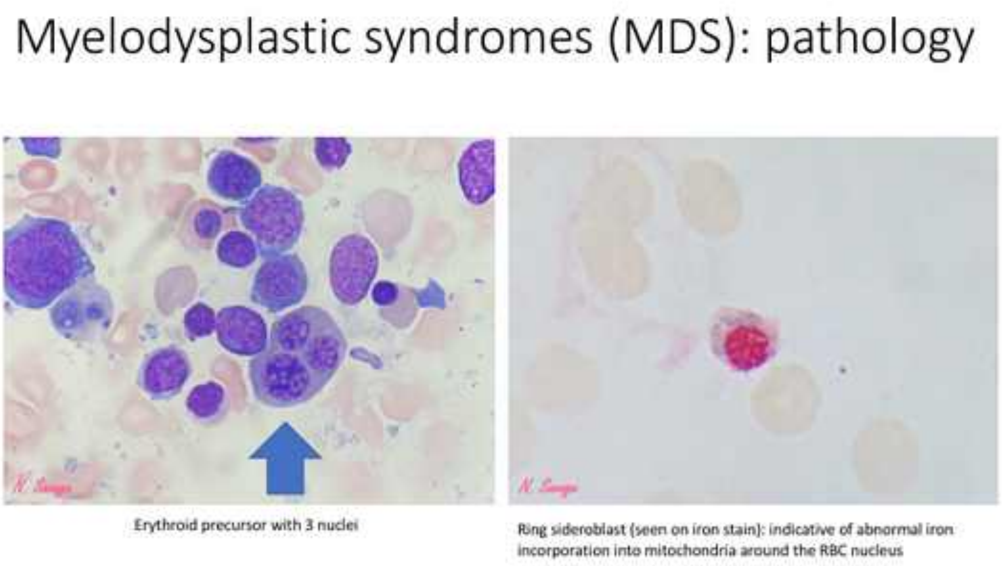
Hypercellular dysfunctional marrow + peripheral cytopenias + dysplasia + risk of progression to AML.

Clinical logic

- More common in older adults.
- Can be found incidentally or present with anemia, infection, or bleeding from cytopenias.
- Blasts remain below the AML threshold; progression to AML reflects clonal evolution and increasing blasts/genetic complexity.
- Risk models in the deck: **IPSS-R** (counts, blasts, cytogenetics) and **IPSS-M** (adds molecular mutations).



Lecture slide 22 - full slide retained. Two morphology anchors: a hypogranular bilobed pseudo-Pelger-Huet neutrophil and abnormally small/dysplastic megakaryocytes.



Lecture slide 23 - full slide retained. The slide shows multinucleated erythroid dysplasia and a ring sideroblast on iron stain, where iron-loaded mitochondria encircle the erythroid nucleus.

COMPANION TREATMENT ENRICHMENT

The companion adds a broad treatment map: supportive transfusion/infection care, hypomethylating agents such as azacitidine/decitabine for selected patients, lenalidomide in del(5q) disease, and allogeneic transplant as the potentially curative approach for suitable higher-risk patients. Treatment is risk-adapted rather than one-size-fits-all.

LO 4 checkpoint

Can you explain how a patient can have a hypercellular marrow yet low circulating blood counts, and identify pseudo-Pelger-Huet cells, micromegakaryocytes, and ring sideroblasts as dysplasia clues?

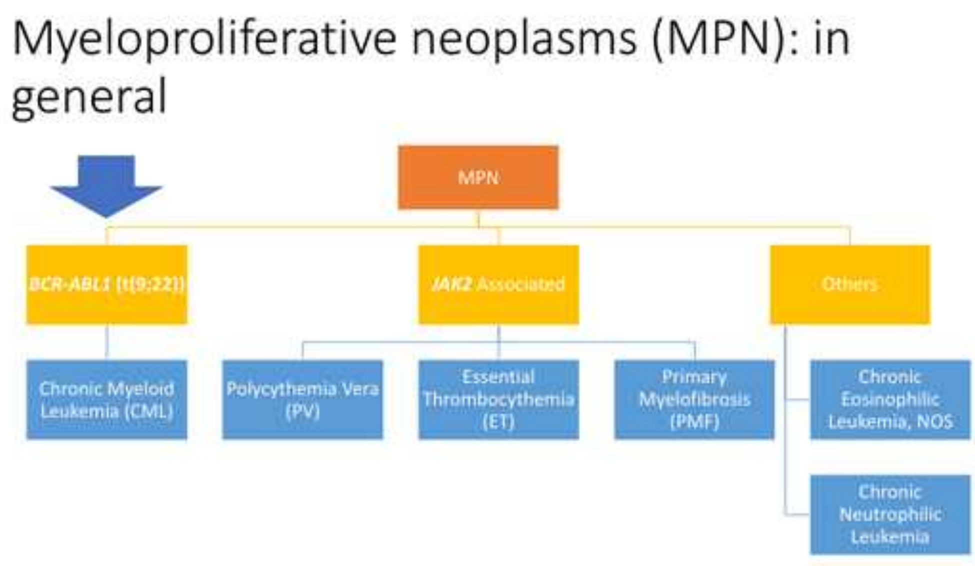
Slides 27-28

Myeloproliferative neoplasms: too many mature cells

LO 1 + LO 6

Slide 27: MPN is the opposite production problem from MDS

- Clonal stem-cell mutations drive production of **too many maturing myeloid cells** that successfully enter blood.
- Often indolent and found in older adults.
- Splenomegaly is common because extramedullary hematopoiesis can develop.
- Any MPN can eventually transform to acute leukemia.



Lecture slide 28 - full slide retained. The deck separates BCR-ABL1-positive CML from the JAK2-associated MPNs PV, ET, and PMF. That split should organize the rest of the lecture.

MPN FORK

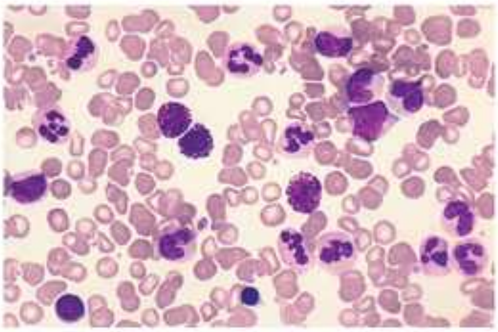
BCR-ABL1 positive -> CML.
BCR-ABL1 negative classic MPNs -> PV, ET, PMF, usually driven by JAK-STAT pathway mutations.

Slides 29-34 CML: BCR-ABL1 and the prototype of targeted therapy LO 5

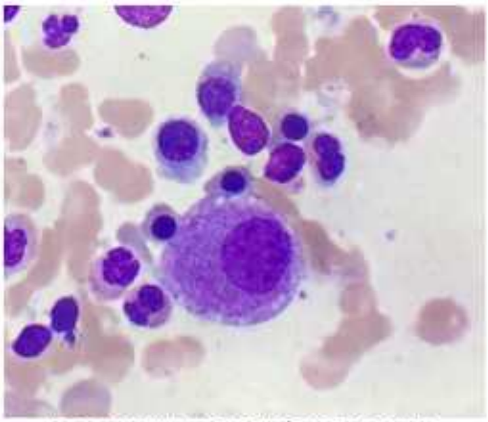
Slides 29-32: recognize the blood picture

- Marked leukocytosis with a **full spectrum of granulocytic maturation / left shift**.
- **Basophilia** is especially important and helps separate CML from benign reactive leukocytosis.
- Marrow is hypercellular with granulocytic proliferation and characteristic small “dwarf” megakaryocytes.
- **LAP score is low** in CML, unlike many leukemoid reactions.

Chronic myeloid leukemia (CML) with *BCR-ABL1*: pathology



Peripheral blood: notice leukocytosis with basophilia

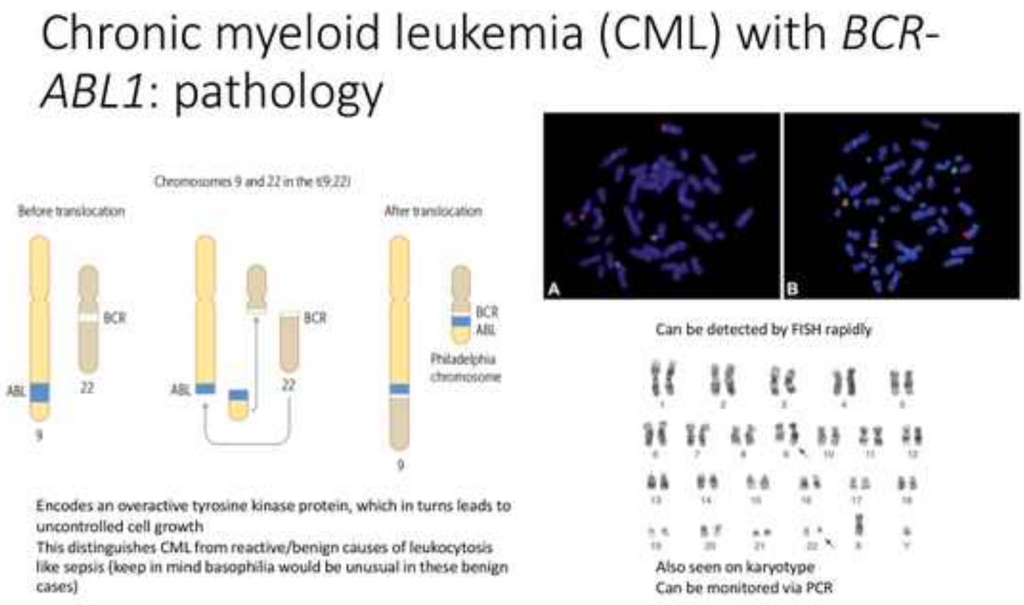


Bone marrow: notice dwarf megakaryocyte

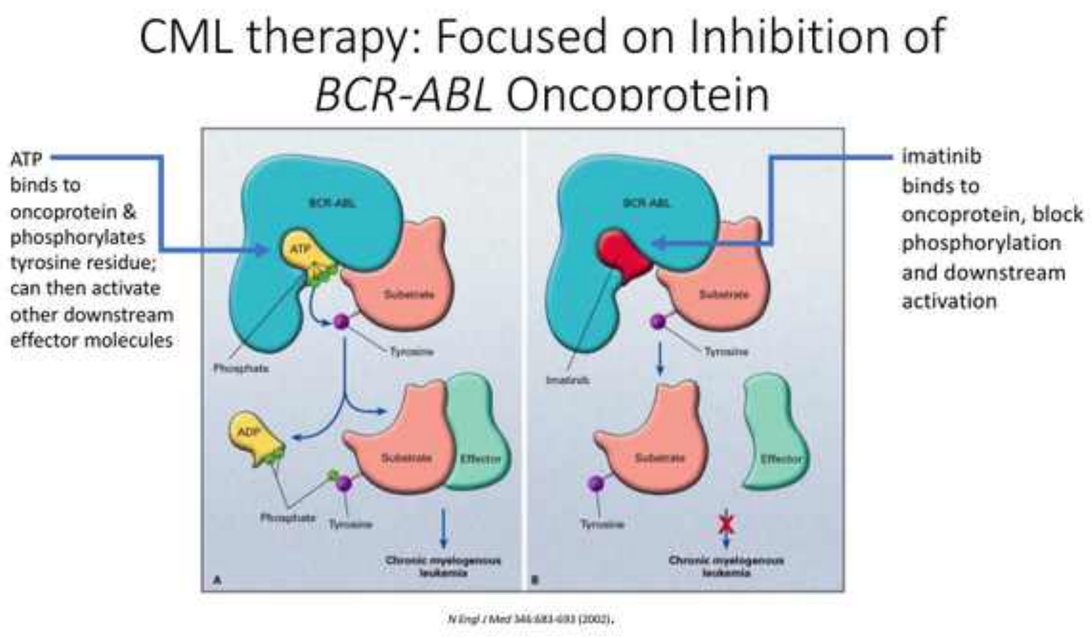
Lecture slide 30 - full slide retained. The blood image demonstrates granulocytic leukocytosis with basophilia; the marrow image shows a small monolobated dwarf megakaryocyte.

CML COLD SET

t(9;22) Philadelphia chromosome -> BCR-ABL1 constitutive tyrosine kinase -> granulocytic leukocytosis + basophilia + low LAP + splenomegaly.



Lecture slide 32 - full slide retained. BCR-ABL1 can be detected by cytogenetics/FISH and monitored quantitatively by PCR. The molecular lesion explains both diagnosis and response monitoring.



Lecture slide 33 - full slide retained. The lecture's targeted-therapy diagram: ATP binding normally permits phosphorylation and downstream signaling; imatinib occupies the kinase and blocks that signaling.

TARGETED THERAPY MEMORY BOX

Ozzy study break - click him for encouragement.

CML changed oncology because the lesion is both diagnostic and druggable: BCR-ABL1 creates the kinase; a TKI such as imatinib directly blocks it.

COMPANION BOARD ENRICHMENT - PHASES

The companion gives the classic progression model: **chronic phase <10% blasts -> accelerated phase 10-19% -> blast phase 20% or more**, with blast crisis behaving as acute leukemia. The current 44-slide deck focuses more on the chronic-phase phenotype and targeted-therapy concept.

CML VS LEUKEMOID REACTION

Both can produce high neutrophils/left shift. **CML favors basophilia, low LAP, and BCR-ABL1.** A reactive infection-associated leukemoid response generally has high LAP and lacks BCR-ABL1.

LO 5 checkpoint

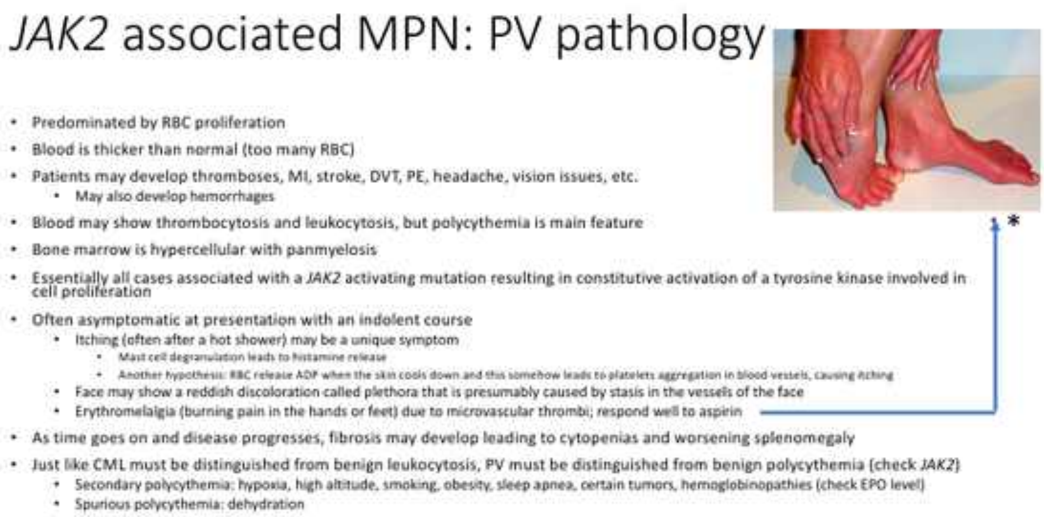
Can you go from t(9;22) -> BCR-ABL1 kinase -> CML blood pattern -> imatinib mechanism, and explain why PCR is useful after diagnosis?

Slides 35-39

JAK-STAT MPNs: PV, ET, and PMF

LO 6

Polycythemia vera (PV)



Lecture slide 36 - full slide retained. PV is dominated by RBC proliferation and panmyelosis. The slide emphasizes thrombosis, aquagenic pruritus, plethora, erythromelalgia, and the need to distinguish true PV from secondary/spurious erythrocytosis.

PV RECOGNITION SET

High Hgb/Hct + JAK2 mutation + low EPO + aquagenic pruritus / erythromelalgia / thrombosis.

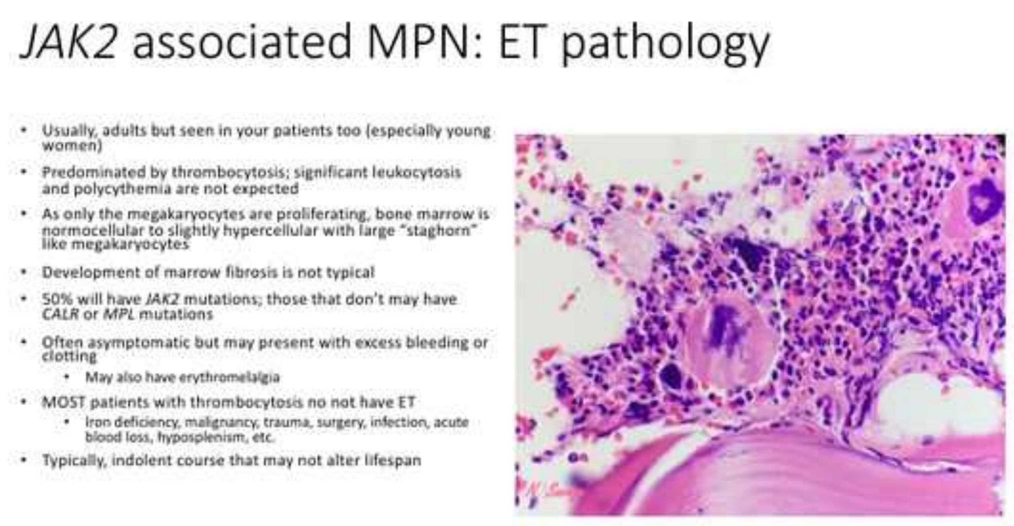
WHY EPO IS LOW

PV is autonomous marrow signaling, so high RBC mass suppresses renal EPO through normal feedback. **Secondary erythrocytosis is driven by EPO and therefore tends to have an elevated/inappropriately normal EPO level.**

COMPANION TREATMENT ENRICHMENT

Recognition-level treatment logic: phlebotomy lowers hematocrit; low-dose aspirin is commonly used for microvascular/thrombotic risk when appropriate; higher-risk patients may require cytoreduction such as hydroxyurea or JAK-pathway therapy.

Essential thrombocythemia (ET)



Lecture slide 37 - full slide retained. ET is dominated by thrombocytosis with large hyperlobulated 'staghorn' megakaryocytes; about half have JAK2, while CALR and MPL account for many JAK2-negative cases.

ET RECOGNITION SET

Persistent thrombocytosis + large mature hyperlobulated megakaryocytes + JAK2/CALR/MPL.

The clinical paradox is **thrombosis OR bleeding**.

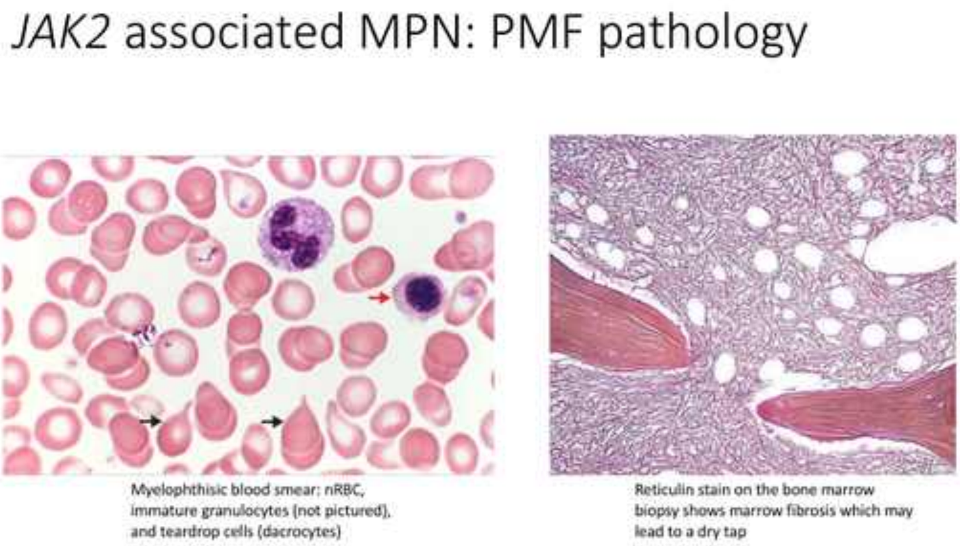
BOARD ENRICHMENT - WHY VERY HIGH PLATELETS CAN BLEED

Extreme thrombocytosis can remove high-molecular-weight vWF multimers from circulation and produce **acquired von Willebrand disease**. So "more platelets" does not always mean "more clotting."

Primary myelofibrosis (PMF)

Slide 38: fibrosis is reactive to the neoplastic clone

- The malignant clone is especially megakaryocytic/granulocytic, but marrow fibrosis develops from **fibroblast-stimulating cytokines released by abnormal megakaryocytes**.
- Progressive fibrosis -> cytopenias, extramedullary hematopoiesis, and worsening splenomegaly.
- Common driver groups: **JAK2, CALR, MPL**.



Lecture slide 39 - full slide retained. The peripheral smear is myelophthisic with nucleated RBCs, immature granulocytes, and teardrop cells. Reticulin staining shows marrow fibrosis that can cause a dry tap.

PMF RECOGNITION SET

Massive splenomegaly + teardrop cells + leukoerythroblastic smear + dry tap + marrow fibrosis + JAK2/CALR/MPL.

	PV	ET	PMF
Dominant cell	RBC / panmyelosis	Platelets / megakaryocytes	Abnormal megakaryocytes + granulocytes, then fibrosis
Drivers	JAK2 essentially all cases in lecture	JAK2, CALR, MPL	JAK2, CALR, MPL
Signature	Low EPO, aquagenic pruritus, thrombosis	Marked thrombocytosis, thrombosis/bleeding	Teardrops, dry tap, massive spleen
Marrow	Panmyelosis	Large hyperlobulated megakaryocytes	Clustered atypical megakaryocytes -> fibrosis

LO 6 checkpoint

Can you identify PV, ET, and PMF from which cell line dominates, their driver mutations, and their signature complication/morphology?

Slides 41-42

MDS/MPN overlap and CMML - for completeness

Beyond official core LO

Slide 42: overlap means dysplasia + proliferation

Savage explicitly includes MDS/MPN overlap “for completeness” and names **chronic myelomonocytic leukemia (CMML)** as the most common example. It is useful as a framework but is not one of the six official Week 7 Myeloid objectives.

COMPANION GAP-FILL - CMML RECOGNITION

- Persistent absolute monocytosis **at least 0.5 x 10^9/L and at least 10% of WBCs**.
- Blasts **<20%**; 20% or more moves into AML territory.
- **No BCR-ABL1** (helps exclude CML).
- Dysplasia in at least one lineage supports the diagnosis.
- Companion separates proliferative vs dysplastic types around a WBC threshold of 13 x 10^9/L.

LOW-PRIORITY BUT USEFUL

CMML = persistent monocytosis + dysplasia + proliferation + no BCR-ABL1 + blasts below AML threshold.

Synthesis

One-table myeloid fingerprint map

LO 1-6

Disease	Core problem / genetics	Blood or morphology clue	Clinical / treatment anchor
AML	Myeloid blasts; often >=20% unless defining genetics	Blasts; Auer rod proves myeloid lineage	Marrow failure; genetics direct risk/ therapy
APL	PML-RARA t(15;17)	Abnormal promyelocytes, many Auer rods	**DIC emergency; ATRA immediately**
MDS	Ineffective dysplastic hematopoiesis	Cytopenias + hypercellular dysplastic marrow; pseudo-Pelger-Huet / ring sideroblast	Older adults; AML progression risk
CML	BCR-ABL1 t(9;22) kinase	Huge granulocytic left shift, basophilia, low LAP	TKI prototype; PCR monitoring
PV	JAK2-driven erythrocytosis	High Hgb/Hct, panmyelosis, low EPO	Aquagenic pruritus, thrombosis, erythromelalgia
ET	JAK2/CALR/MPL megakaryocytic proliferation	Very high platelets, large hyperlobulated megakaryocytes	Thrombosis or bleeding; acquired vWD if extreme
PMF	JAK2/CALR/MPL clone -> cytokine-driven fibrosis	Teardrops, leukoerythroblastosis, dry tap	Massive splenomegaly / extramedullary hematopoiesis
CMML	MDS/MPN overlap	Persistent monocytosis + dysplasia	Blasts <20%, BCR-ABL1 absent

MUTATION MAP TO MEMORIZE

APL = PML-RARA (15;17).
CML = BCR-ABL1 (9;22).
PV = JAK2.
ET / PMF = JAK2, CALR, or MPL.

COMMON EXAM TRAPS

- **Very high WBC does not automatically mean AML.** CML can have WBC >100K while cells are mostly maturing granulocytes.
- **Basophilia strongly favors CML** over a simple leukemoid response.
- **Low EPO favors PV;** high EPO favors secondary erythrocytosis.
- **Teardrops + dry tap** point toward marrow fibrosis, not ET.
- **APL is an emergency before FISH confirmation.**
- **MDS has cytopenias despite a busy marrow.**

Learning-Objective Synthesis

Official LO	What you should be able to do without notes
LO 1	Distinguish acute blast-rich disease from chronic mature-cell disease and organize AML vs MDS vs MPN vs overlap.
LO 2	Recognize an AML blast/Auer rod, explain marrow failure, and know why recurrent genetics matter without memorizing every WHO/ICC category.
LO 3	State PML-RARA, DIC risk, ATRA mechanism, urgency, and differentiation syndrome.
LO 4	Explain MDS ineffective hematopoiesis and recognize characteristic dysplasia/cytopenias and AML-progression risk.
LO 5	Go from Philadelphia chromosome to BCR-ABL1 kinase to CML phenotype to imatinib/TKI targeted therapy and PCR monitoring.
LO 6	Separate PV, ET, and PMF by dominant cell line, JAK-STAT driver, blood/marrow findings, and major complications.

LAST 60 SECONDS

AML: blasts / Auer rod / marrow failure.

APL: 15;17 PML-RARA + DIC -> ATRA now.

MDS: hypercellular dysfunctional marrow + cytopenias + dysplasia.

CML: 9;22 BCR-ABL1 + basophilia + low LAP -> TKI.

PV: JAK2 + high Hct + low EPO + itching/thrombosis.

ET: JAK2/CALR/MPL + platelets + thrombosis/bleeding.

PMF: JAK2/CALR/MPL + teardrops + dry tap + massive spleen.

Source hierarchy: Savage's 44-slide Myeloid Neoplasms deck is the primary slide-order source. The official Week 7 objectives control emphasis. The uploaded Lecture 4 and 5 companion is used as supplemental explanation for broad treatment logic, CML phases, CMML, and classic board distinctions. Detailed classification tables explicitly marked “No need to memorize” are not turned into artificial memorization burden.